

# Spine Large-Graph Campaign Results

Correctness-Gated Spine vs. GraSU+ReGraph K1/K4-shared

Chuxiao Han

Live campaign snapshot, July 2026

## Abstract

This report is generated directly from correctness-gated campaign artifacts. The snapshot is **PARTIAL**: 70 of 93 runnable physical executions are currently admitted, with 2 of 95 contract cases excluded by proven physical capacity bounds and 19 complete three-system groups. Missing bars are unexecuted comparisons, not zero-valued measurements. Every plotted row passes exact integer-state comparison or the declared  $10^{-5}$  float32 PageRank output tolerance. Headline plots use insertion batch 8; the update sweep labels operation and batch size explicitly.

## 1 End-to-end performance

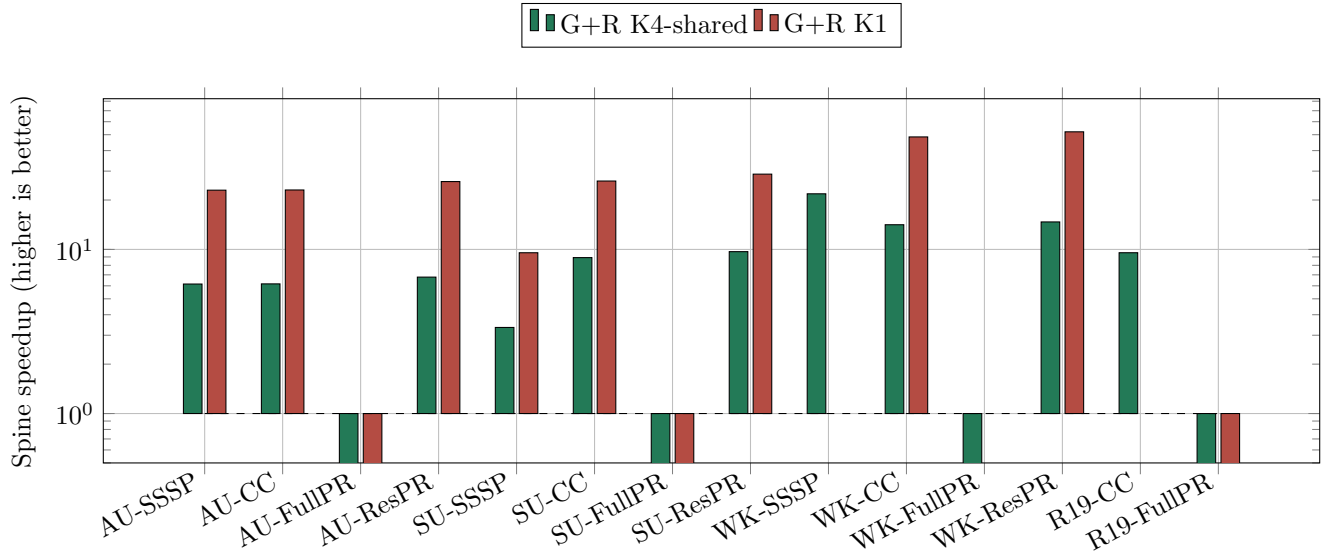


Figure 1: Device-cycle speedup after both architecture and independent mathematical correctness gates. K4-shared is the primary GraSU baseline.

## 2 Memory traffic and HBM energy

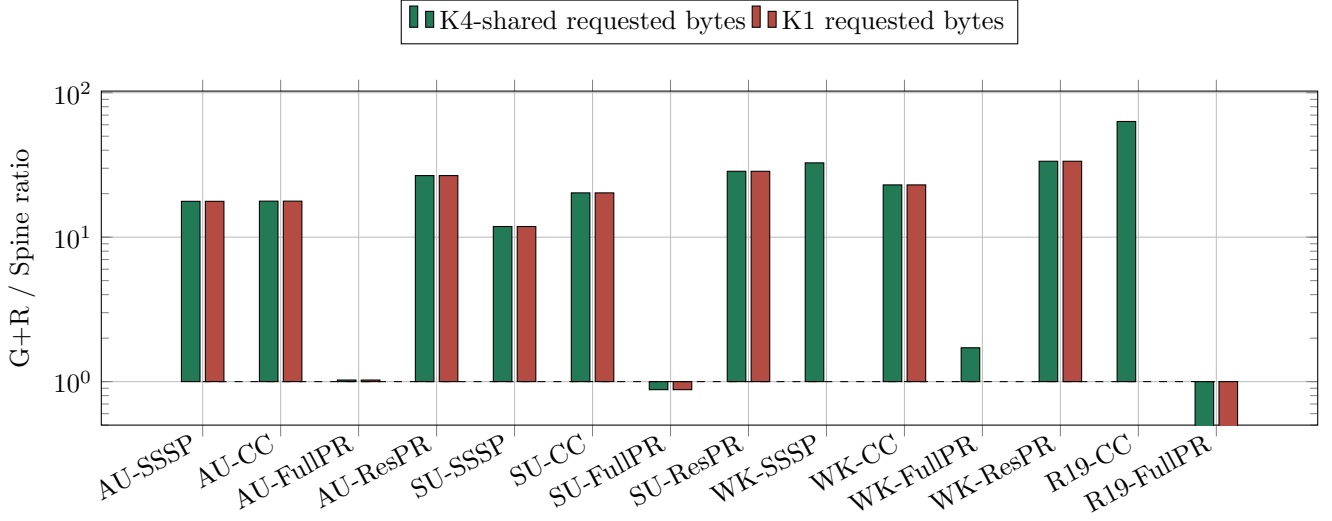


Figure 2: Accepted backend bytes. Ratios above one mean GraSU+ReGraph moves more data than Spine; request and DRAM completion ledgers close on every row.

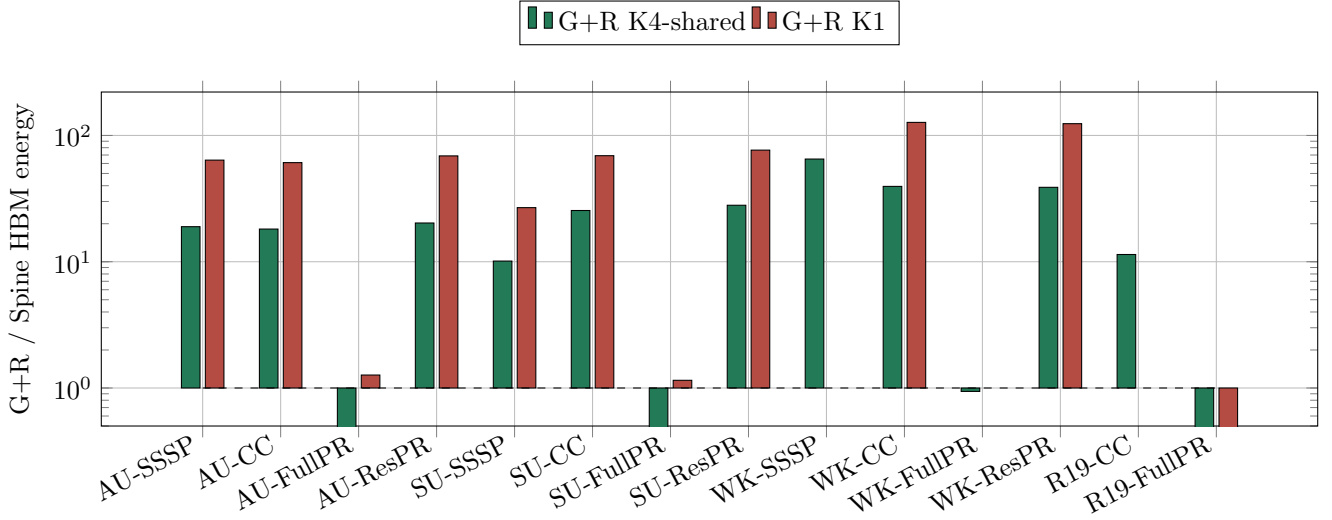


Figure 3: All-controller DRAMSim3 energy ratio. This is HBM energy, not total accelerator or board energy.

### 3 Update-only tradeoff

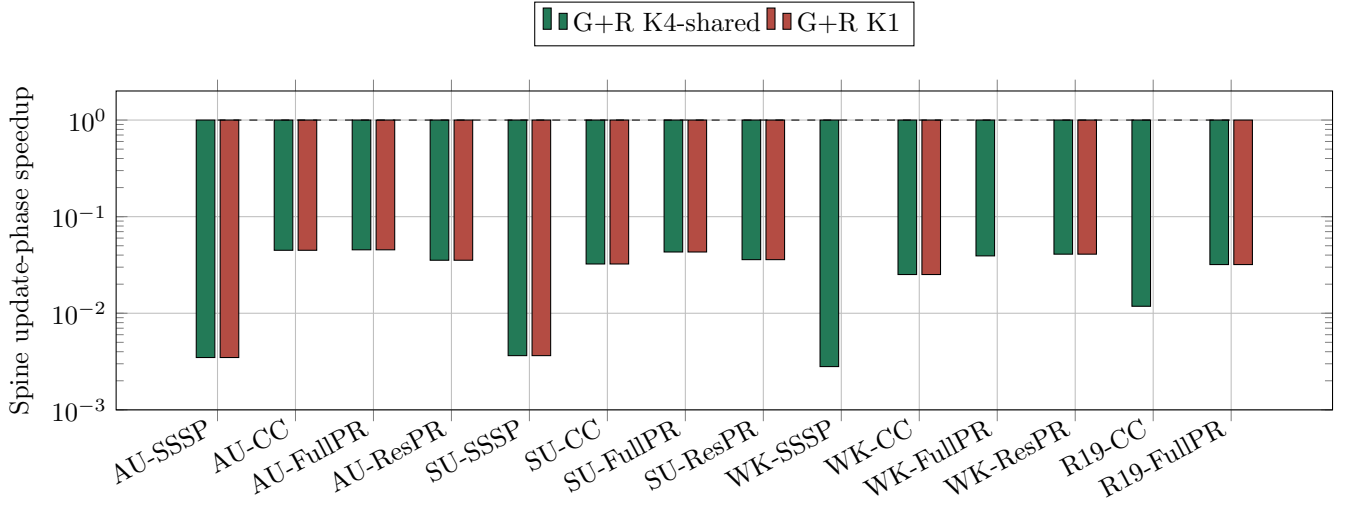


Figure 4: Update-phase cycles only. Values below one expose Spine’s current maintenance cost even when end-to-end execution favors Spine.

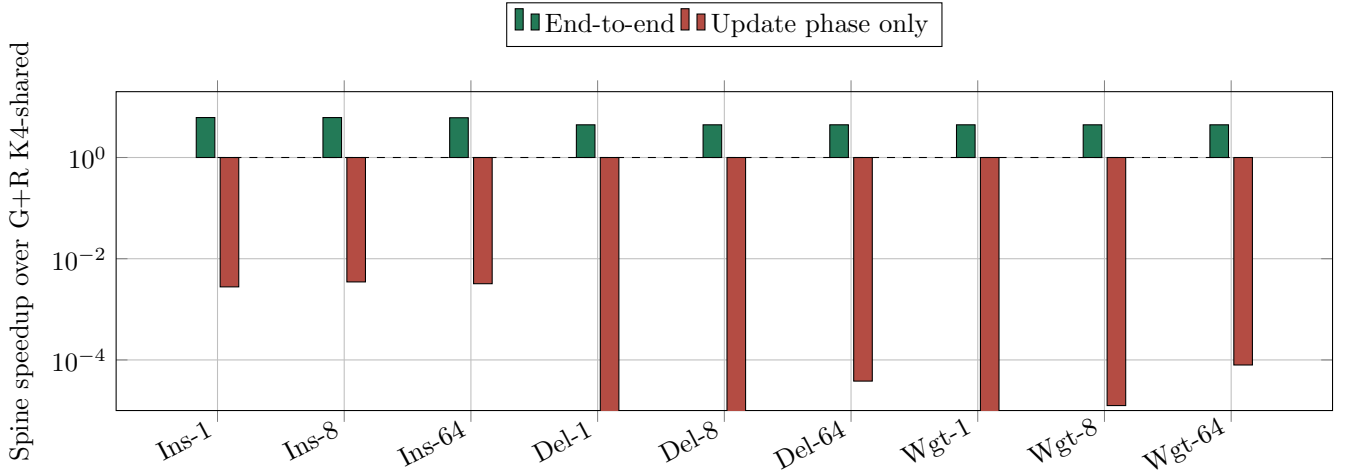


Figure 5: AskUbuntu weighted-SSSP operation and batch sweep. Ins, Del, and Wgt denote insertion, deletion, and weight change. Update speedup uses semantic user mutations and the isolated update-phase cycle window.

## 4 Dense-batch behavior

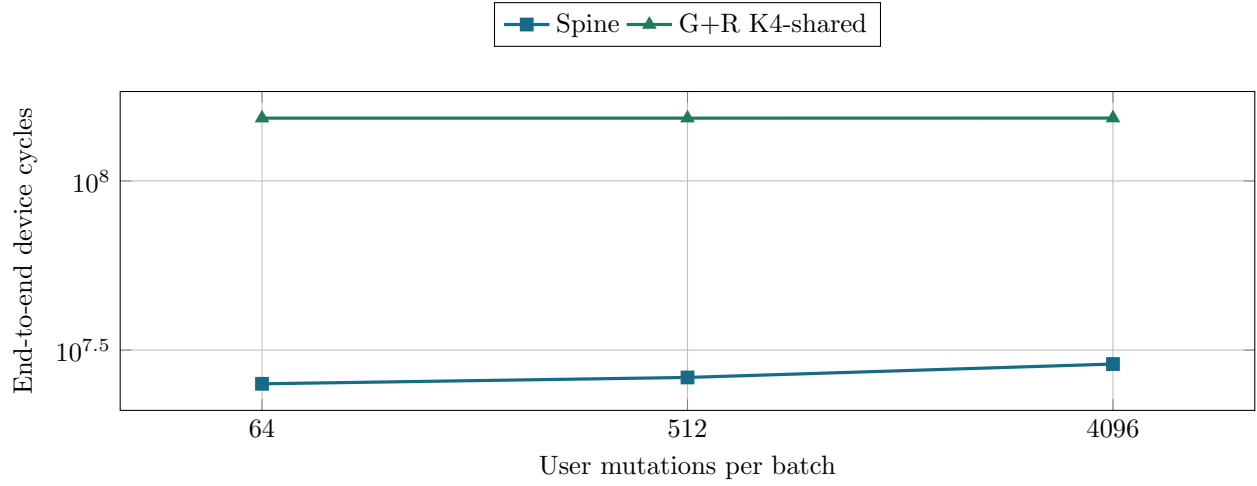


Figure 6: Correctness-gated AskUbuntu weighted-SSSP insertion latency as the batch grows from 64 toward 4096 mutations. Missing points are still running, not zero-valued measurements.

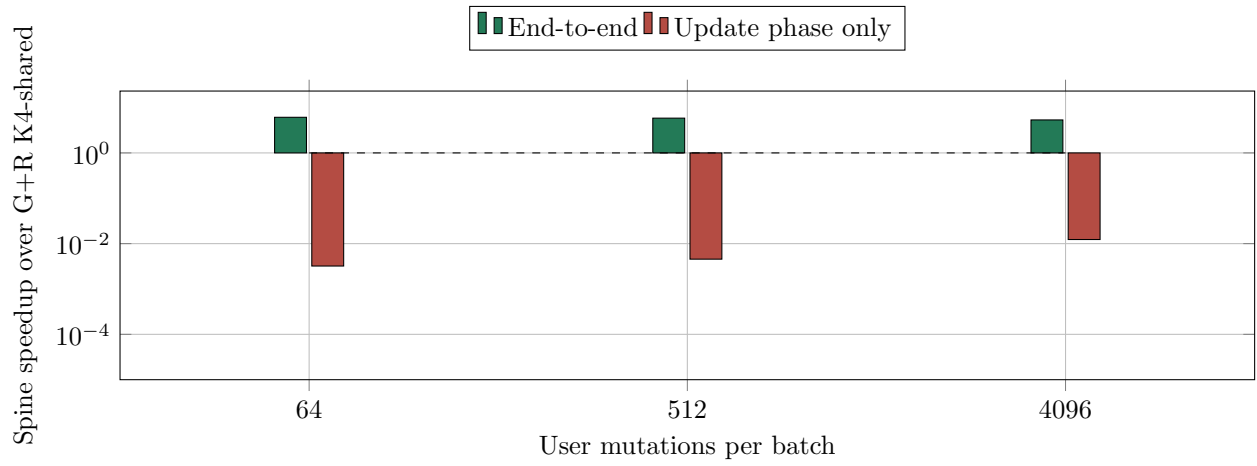


Figure 7: Dense-batch tradeoff. End-to-end and isolated update-phase windows are kept separate because the current Spine maintenance path can lose even when its complete execution wins.

## 5 Implementation feasibility and component power

| Build             | LUT    | FF     | BRAM | URAM | DSP | WNS ns | Setup         |
|-------------------|--------|--------|------|------|-----|--------|---------------|
| Spine opt-v2 SSSP | 136437 | 159403 | 95   | 115  | 18  | 0.003  | closed        |
| G+R SSSP          | 96559  | 107380 | 233  | 64   | 0   | -0.013 | target missed |
| G+R Full PR       | 178374 | 177836 | 278  | 64   | 304 | -0.130 | target missed |
| G+R Residual PR   | 247692 | 301029 | 293  | 64   | 336 | -0.087 | target missed |

Table 1: Absolute U55C routed resources and timing. Spine is a native SSSP build; GraSU+ReGraph rows are algorithm-specific conversion-free whole systems. They establish feasibility but are not an iso-functional area ratio.

| Build             | Dynamic W | HBM sub. | Update | Compute | FIFO  | Other |
|-------------------|-----------|----------|--------|---------|-------|-------|
| Spine opt-v2 SSSP | 27.677    | 17.674   | 1.891  | 0.205   | 0.003 | 0.113 |
| G+R SSSP          | 21.921    | 10.807   | 2.322  | 0.777   | 0.035 | 0.177 |
| G+R Full PR       | 25.104    | 12.685   | 1.202  | 3.133   | 0.079 | 0.189 |
| G+R Residual PR   | 26.014    | 12.372   | 1.957  | 3.589   | 0.068 | 0.186 |

Table 2: Vivado vectorless hierarchy attribution with Low confidence. These values identify implementation components; they are not workload energy or board power. Workload-specific HBM energy in Figures 3–4 comes independently from DRAMSim3 command and background activity.

**Workload-specific component activity.** The correctness-gated snapshot exports 464 component rows covering maintenance/update, reader/source cache, compute/gather/apply, streams, selected on-chip arrays, and the HBM frontend. These are physical activity counters with explicit per-component contracts, not energy estimates; they are archived in `component_activity.csv` for subsequent calibrated per-event energy analysis.

## 6 Correctness coverage

| Algorithm | Cross-system groups | Exact | K1/K4 triplets | Failed |
|-----------|---------------------|-------|----------------|--------|
| SSSP      | 13                  | 13    | 10             | 0      |
| CC        | 4                   | 4     | 3              | 0      |
| FullPR    | 4                   | 0     | 3              | 0      |
| ResPR     | 3                   | 3     | 3              | 0      |

Table 3: Only groups containing at least two systems are counted. Exact means canonical final-state equality after both architecture and independent mathematical-oracle admission.

## 7 Frozen dataset catalog

| Data | Vertices | Directed edges | CC edges  | ResPR edges | Spine full |
|------|----------|----------------|-----------|-------------|------------|
| AU   | 515281   | 390847         | 679246    | 826848      | yes        |
| SU   | 567316   | 616118         | 1061248   | 1075035     | yes        |
| WK   | 1140149  | 1142352        | 2032558   | 2190639     | yes        |
| SO   | 6024271  | 23724166       | 41008044  | 28299669    | yes        |
| BC   | 24575382 | 38968503       | 77937006  | 52778301    | no         |
| HW   | 1139905  | 112751422      | 112751422 | 112784084   | yes        |
| PK   | 1632803  | 44603928       | 44603928  | 44603928    | yes        |
| OK   | 3072441  | 234370166      | 234370166 | 234370166   | yes        |
| LJ   | 4847571  | 85702474       | 85702474  | 85703436    | yes        |
| LJ08 | 5363260  | 99028542       | 99028542  | 99028616    | yes        |
| UK   | 18520486 | 292243663      | 523574516 | 295209230   | no         |
| R19  | 524288   | 15483485       | 29732038  | 15672181    | yes        |

Table 4: Immutable materialized graph views. Full PageRank uses the separately frozen 4M-edge cap. “Spine full” is the current  $2^{24}$ -vertex admission check, not a performance result.

## 8 Simulator engineering runtime

| System        | Completed runs | Median host s | Maximum host s |
|---------------|----------------|---------------|----------------|
| Spine         | 16             | 644.4         | 13263.7        |
| G+R K4-shared | 15             | 2280.3        | 23318.1        |
| G+R K1        | 11             | 5867.6        | 27529.1        |

Table 5: Observed host wall time under concurrent campaign load. These values measure simulator throughput and are not accelerator performance.

**Claim boundary.** Only complete correctness-gated pairs enter the figures. Current FPGA/RTL evidence supports feasibility and PPA separately; this report does not claim cycle-for-cycle FPGA calibration or ASIC total power.